

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [109, 106, 108, 113, 107, 132, 109, 88, 99, 102]

Tap amplitudes (pixels): [175.5135322418189, 195.68597292601225, 183.19934497699495, 167.26326554267678, 189.7893569197177, 151.21177202850313, 165.5294535724685, 190.19200824430033, 151.21177202850313, 184.29595763336752, 180.47160441465576]

Gemini 3 Pro

To score the subject on MDS-UPDRS Item 3.4: FINGER TAPPING, let's analyze the provided features against the scoring criteria.

Analysis of Extracted Features

1. Speed and Rhythm (Tap Periods):

- **Data:** [109, 106, 108, 113, 107, 132, 109, 88, 99, 102]
- **Observation:** The periods are relatively consistent, hovering around 100-110 frames.
 - There is a slight spike at the 6th period (132 frames), which indicates a minor hesitation or slowing, but it recovers immediately.
 - There is a dip at the 8th period (88 frames), indicating a faster tap, followed by a return to baseline.
- **Conclusion:** There are no significant interruptions, freezes, or severe slowing. The rhythm is largely regular with only very minor fluctuations that do not constitute distinct "interruptions" or "hesitations" sufficient for a higher score.

2. Amplitude (Tap Amplitudes):

- **Data:** [175.5, 195.7, 183.2, 167.3, 189.8, 151.2, 165.5, 190.2, 151.2, 184.3, 180.5]
- **Observation:**
 - The amplitudes fluctuate but remain generally high (mostly between 150 and 195 pixels).
 - There is no consistent *decrementing* trend. For example, the amplitude drops at tap 6 (151.2) but bounces back up at tap 8 (190.2) and ends strong at tap 11 (180.5).

- **Conclusion:** The criteria for scores 1, 2, and 3 specifically look for amplitude *decrementing* (getting smaller and smaller) at specific points in the sequence (near the end, midway, or after the 1st tap). This subject maintains amplitude throughout the sequence without a sustained decrement.

Scoring Determination

- **Score 0 (Normal):** Requires "No problems."
- **Score 1 (Slight):** Requires one or two interruptions, slight slowing, or amplitude decrement near the end.
 - The period variation (132 vs 109) is very minor and likely within normal variance rather than a pathological hesitation.
 - There is no amplitude decrement near the end.
- **Score 2, 3, 4:** The subject clearly does not meet the criteria for these higher scores (no multiple interruptions, no mild/moderate/severe slowing, no early amplitude decrement).

The performance shows consistent speed and amplitude without significant hesitations or decrementing.

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